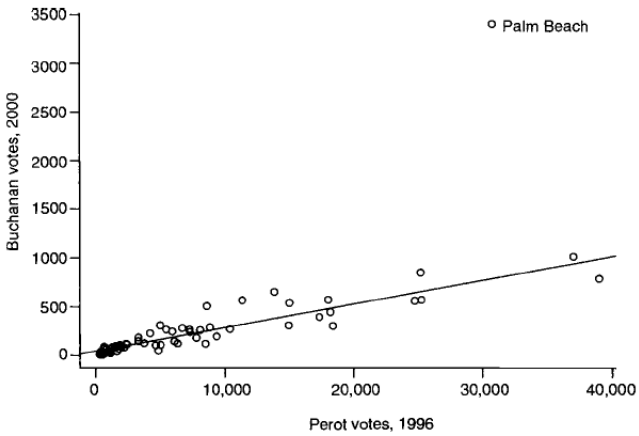
1 Linear Probability Model I

In the 2000 U.S. presidential election, Palm Beach County in Florida was the focus of unusual voting patterns apparently caused by a confusing "butterfly ballot." Many voters claimed that they voted mistakenly for the Reform Party candidate, Pat Buchanan, when they intended to vote for Al Gore. Figure 4.7 shows the total number of votes for Buchanan plotted against the number of votes for the Reform Party candidate in 1996 (Ross Perot), by county in Florida.



a) In county i, let π_i, denote the proportion of the vote for Buchanan and let x_i, denote the proportion of the vote for Perot in 1996. For the linear probability model fitted to all counties except Palm Beach, $\pi_i = -0.0003 + 0.0304x_i$. Give the value of P in the interpretation: The estimated proportion vote for Buchanan in 2000 was roughly P% of that for Perot in 1996.

In the equation,

$$\pi_i = -0.0003 + 0.0304x_i \tag{1.1}$$

the intercept is $x_o = -0.0003 \approx 0$. Hence,

$$\pi_i \approx 0.0304x_i$$

The estimated proportion vote for Buchanan in 2000 was roughly 3.04% of that for Perot in 1996. We no longer consider percentage points due to the small baseline, reducing the relationship into a direct proportion.

b) For Palm Beach County, $\pi_i = 0.0079$ (Buchanan) and $x_i = 0.0774$ (Perot). Does this result appear to be an outlier? Investigate, by finding $\pi_i/\hat{\pi}_i$. (George W. Bush won the state by 537 votes and, with it, the Electoral College and the election. Other ballot design problems played a role in 110,000 disqualified "overvote" ballots, in which people mistakenly voted for more than one candidate, with Gore marked on 84,197 ballots and Bush on 37,731.

Using $x_i = 0.0774$ as input to Equation (1.1), we get 0.00205. This is the expected proportion of votes for Buchanan if voting behavior is similar to all counties except Palm Beach. This gives $\pi_i/\hat{\pi}_i = 3.8481$. This states that the proportion of votes for Buchanan is 3.8481 times higher than the expected and this may have been influenced by confusing "butterfly ballot."

2 Linear Probability Model II

For the data on snoring and heart disease, refit the linear probability model or the logistic regression model using the scores (a) (0, 2 ,4, 6), (b) (0, 1, 2, 3), and (c) (1, 2, 3, 4). Compare $\hat{\beta}$ for the three choices. Compare fitted values. Summarize the effect of linear transformations of scores, which preserve relative sizes of spacings between scores.

Table 1: Linear Probability Model Coefficients across Scores

	a	b	c	d
$\hat{\beta}_0$	0.01760	0.01760	-0.01864	0.01760
$\hat{\beta}_1$	0.01812	0.03625	0.03625	0.00072

Note:
I added d with scores 0, 50, 100, 150.

Table 2: Linear Probability Model Fitted Values across Scores

a		b		c		d	
Scoring	Fitted	Scoring	Fitted	Scoring	Fitted	Scoring	Fitted
0	0.01760	0	0.01760	1	0.01760	0	0.01760
2	0.05385	1	0.05385	2	0.05385	50	0.05385
4	0.09009	2	0.09009	3	0.09009	100	0.09009
6	0.12634	3	0.12634	4	0.12634	150	0.12634

Preserving the spacing between chosen score representations yields different β_0's unless the same baseline value is used - as shown in Table 1. $\hat{\beta}_{1(s=1)}$ with 1 spacing ($s = 1$) in between scores can be used to calculate $\hat{\beta}_1$ when $s = m > 1$ by using $\hat{\beta}_{1(s=1)}/s$. It results in the same predictions or fitted values $\hat{\pi}_i$ for each score as illustrated in Table 2.

3 Logistic Regression

Denote by M_0 the model $logit[P(Y = 1)] = \alpha$ and by M_1 the model $logit[P(Y = 1)] = \alpha + \beta x$. Denote the maximized log-likelihood values by L_0 for M_0, L_1 for M_1, and L_s for the saturated model. Create a data file in two ways, entering the data as (i) ungrouped data: $n_i = 1, i = 1, \ldots, 12$; and (ii) grouped data: $n_i = 4, i = 1, 2, 3$.

Table 3: Ungrouped Data

x	y	x	y
0	1	1	0
0	0	1	0
0	0	2	1
0	0	2	1
1	1	2	1
1	1	2	1

Table 4: Grouped Data

x	0	1	totals
0	3	1	4
1	2	2	4
2	0	4	4

a) Fit M_0 and M_1 for each data file. Report L_0 and L_1 in each case. Note they are the same for each form of data entry.

```
M0_data <- glm(
  data$y ~ 1,
  family=binomial(link="logit"))

M0_grouped_data <- glm(
  grouped_data$`1`/grouped_data$totals ~ 1,
  family=binomial(link="logit"),
  weights = grouped_data$totals)

M1_data <- glm(
  data$y ~ data$x,
  family=binomial(link="logit"))

M1_grouped_data <- glm(
  grouped_data$`1`/grouped_data$totals ~ grouped_data$x,
  family=binomial(link="logit"),
  weights = grouped_data$totals)
```

Table 5: Logistic Regression Models Fitted Values across Data Formats

	M_0		M_1	
	Ungrouped	Grouped	Ungrouped	Grouped
$\hat{\beta}_0$	0.33647	0.33647	-1.5027	-1.5027
$\hat{\beta}_1$			2.0605	2.0605

```
calculate_loglikehood <- function(k, pi_hat, n){
  sum(k * log(pi_hat) + (n - k) * log(1 - pi_hat))
}

L0_data <- calculate_loglikehood(k = data$y,
                                 pi_hat = predict(M0_data, type="response"),
                                 n = rep(1,12))

L0_grouped_data <- calculate_loglikehood(k = grouped_data$`1`,
                                         pi_hat = predict(
```

```
                                          M0_grouped_data, type="response"),
                                        n = grouped_data$totals)

L1_data <- calculate_loglikehood(k = data$y,
                                 pi_hat = predict(M1_data, type="response"),
                                 n = rep(1,12))

L1_grouped_data <- calculate_loglikehood(k = grouped_data$`1`,
                                         pi_hat = predict(
                                          M1_grouped_data, type="response"),
                                         n = grouped_data$totals)
```

Table 6: LogLikelihood Values

L_0		L_1	
Ungrouped	Grouped	Ungrouped	Grouped
-8.1503	-8.1503	-5.5141	-5.5141

b) Show that the deviances for M_0 and M_1 differ for the two forms of data entry. Why is this? [Hint: How many parameters are in the saturated model for each data file?]

The deviances in Table 7 differ for both M_0 and M_1 despite data only differing by form. This is because the form changes the degrees of freedom. Grouped data has 3 rows and 2 columns $df = (3-1)(2-1)$ while ungrouped has 12 rows and 2 columns $df = (12-1)(2-1)$

Table 7: Deviance and DF Values

	M_0		M_1	
	Ungrouped	Grouped	Ungrouped	Grouped
deviance	16.301	6.2568	11.028	0.9844
DF	11.000	2.0000	11.000	2.0000

c) Show that the difference between the deviances for M_0 and M_1 is the same for each form of data entry. Why is this? (Thus, for testing the effect of x, it does not matter how you enter the data, but it does matter if you want to test goodness of fit.)

Regardless of data format, following Equation (3.1), and showing the loglikelihood values in Table 6, for both the grouped and ungrouped data, $\mathcal{D}_0 - \mathcal{D}_1 = 5.27238$.

$$\begin{aligned} \mathcal{D}_\prime - \mathcal{D}_1 &= -2(L_0 - L_s) - [-2(L_1 - L_s)] \\ &= -2L_0 + 2L_s + 2L_1 - 2L_s \\ &= -2L_0 + 2L_1 \end{aligned} \tag{3.1}$$

4 Poisson Loglinear Model

The table below shows the SAS output for a Poisson loglinear model fit using `x = weight` and `Y = number of satellites`.

Criterion	DF	Value
Deviance	171	560.8664
Pearson Chi-Square	171	535.8957
Log Likelihood		71.9524

Parameter	Estimate	Std Error	Wald 95%	Conf Limits	Chi-Sq	Pr > ChiSq
Intercept	−0.4284	0.1789	−0.7791	−0.0777	5.73	0.0167
weight	0.5893	0.0650	0.4619	0.7167	82.15	< .0001

a) Use $\hat{\beta}$ to describe the weight effect. Show how to construct the reported confidence interval.

- Weight effect: A 1-unit increase in weight has a multiplicative impact of $e^{0.5893} = 1.80273$ in the estimated number of satellites.

- Confidence interval:
  - LB: `(0.5893 - 1.96*0.065) =` 0.4619
  - UB: `(0.5893 + 1.96*0.065) =` 0.7167

b) Construct a Wald test that Y is independent of x. Interpret. What else do you need to conduct a likelihood-ratio test of this hypothesis?

To test if Y and x are independent, we can set $H_0 : \beta_1 = 0$ and use Wald test statistic.

$$\frac{(\hat{\beta}_1 - \beta_1)^2}{Var(\hat{\beta}_1)} = \frac{(0.5893 - 0)^2}{0.065^2} = 82.19515 > 3.8416$$

We are 95% confident that if we repeatedly collect sample and estimate β_1, its value will be found within the lower and upper bounds, and this does not include 0. We reject independence hypothesis.

To conduct a likelihood-ratio test of this hypothesis, I need to fit a model similar to M_0 in the previous item where we only have the intercept. If the deviance of the current model and that of M_0 is just the same, there is evidence towards independence.

c) Is there evidence of overdispersion? Explain.

Overdispersion is a phenomenon that sometimes occurs in data that are modeled with the binomial or Poisson distributions. If the estimate of dispersion after fitting, as measured by the deviance or Pearson's chi-square, divided by the degrees of freedom, is not near 1, then the data might be overdispersed if the dispersion estimate is greater than 1 or underdispersed if the dispersion estimate is less than 1 (See SAS documentation). — Since the value shown below is greater but close to 1, there may be no overdispersion.

$$\frac{\text{deviance}}{\text{DF}} = \frac{560.8664}{171} = 3.27992$$